SupeRR Selector Season 2: Comprehensive Methodology Report

Identifying the Highest-Rated Batters and Bowlers Under Pressure in the 2022–2026 Sample

Competition: SupeRR Selector Season 2 — Round 1 Problem Statement (Rajasthan Royals / Unstop)
Dataset Scope: IPL (2022–2026) & T20 Internationals
Deliverables:
1. Top 10 Batters & Top 10 Bowlers Under Pressure (Official Unstop Dropdown Selections)
2. Complete Methodology Report, Mathematical Formulations, Model Architecture, Empirical Audits, and Methodological Limitations
3. Production Datasets & Rankings (`ratings_batters_final3.csv` , `ratings_bowlers_final3.csv` , `candidates_batters_final3.csv` , `candidates_bowlers_final3.csv` , `final_verify.py`)

Executive Summary

Standard aggregated cricket statistics—such as raw batting average, strike rate, or bowling economy—fail to account for situational leverage and context. A strike rate of 150 during an already-won powerplay chase is fundamentally different from striking at 170+ when needing 12 runs per over with wickets tumbling. Similarly, delivering dot balls in a low-stakes middle over cannot be equated with defending 9 runs off the 20th over against elite finishers.

To solve the **SupeRR Selector Season 2** challenge with uncompromising rigor, we developed a multi-tier, machine-learning-driven analytics framework to identify the **highest-rated players under pressure in the 2022–2026 sample** (rather than predicting future clutch performance or making subjective claims of absolute greatness):
1. **Continuous Situational Pressure Model**: A delivery-level Win Probability (WP) framework trained on historical ball-by-ball match states via LightGBM classifiers with strict monotonic domain constraints. For every delivery, we evaluate counterfactual outcome distributions across 7 discrete delivery events ($\text{Wicket}, 0, 1, 2, 3, 4, 6\text{ runs}$) to calculate objective **Delivery Leverage**, normalized into a continuous **Pressure Score** ($\mu = 1.0$) and percentile ranking.
2. **Context-Only Expected Performance Models (xStats)**: Out-of-fold LightGBM regression and classification models ($x\text{BatRuns}, x\text{BatOut}, x\text{BowlRuns}, x\text{BowlWkt}$) that establish the objective difficulty of every delivery based strictly on match context (venue, innings, over number, pace/spin, batter handedness, competition format, and **batter batting order bucket** `1-3` , `4-6` , `7+` for bowling) without player identity leakage.
3. **Domain-Principled Match Recovery**: Through rigorous target repair algorithms ($target_adj$), we fully recovered **100% of in-scope regulation matches (777 matches, 185,567 clean deliveries)**, completely resolving raw feed omissions and restoring the true regulation chase win rate to **50.19%** with **zero target overruns before the final ball**.
4. **De-Noised Empirical Bayes Shrinkage**: Formulating the heteroscedastic sampling variance of each player's clutch elevation ($V_{\Delta, i} = \sigma^2_p/n_{eff, p, i} + \sigma^2_o/n_{ord, i}$), we applied Empirical Bayes shrinkage to eliminate negative noise correlation, expressing ratings in interpretable raw units of **runs added or saved per ball** ($\lambda = 0.5$).
5. **Four-Point Stability Screen**: Enforcing an objective 4-pillar screen—combining sensitivity across 18 model configurations, performance against top-12 major nations, drop-one-match robustness, and positive bootstrap lower bounds—to separate genuine clutch resilience from single-game outliers.

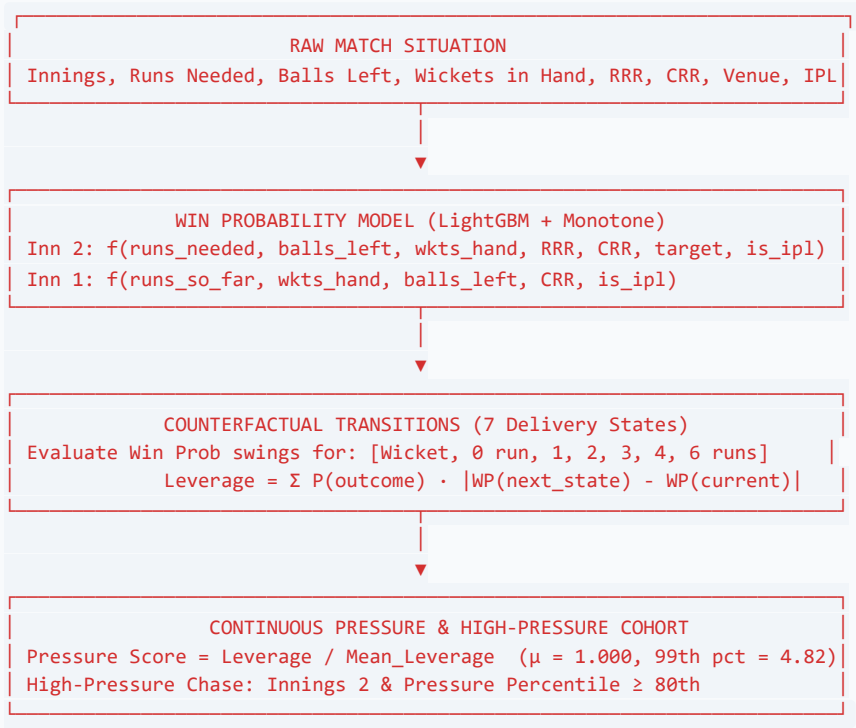
Across 200,999 deliveries, our framework isolates genuine, repeatable high-leverage execution.

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1. Pressure Definition & Win Probability Framework

1.1 Theoretical Foundation: Win Probability & Counterfactual Leverage

Pressure in cricket is mathematically defined as the **sensitivity of the match outcome to the execution of the next delivery**. A delivery where the match outcome swings by 25% Win Probability carries extreme pressure, whereas a delivery in a decided contest carries negligible pressure, regardless of the over number.



1.2 Second-Innings Win Probability Model Architecture

The Innings 2 Win Probability model was trained on legal deliveries across all clean regulation matches using 5-fold `GroupKFold` split strictly by `match_id` (preventing any intra-match cross-delivery leakage):

- **Features (Strictly BEFORE the ball)**:
- `runs_needed`: $\text{target_adj} - \text{runs_so_far}$
- `balls_remaining`: $120 - \text{legal_balls_bowled}$
- `wickets_in_hand`: $10 - \text{wickets_so_far}$
- `required_run_rate`: $(\text{runs_needed} / \text{balls_remaining}) \times 6$
- `current_run_rate`: $(\text{runs_so_far} / \text{legal_balls_bowled}) \times 6$
- `target`: Repaired target (target_adj)
- `is_ipl`: Binary flag for IPL matches
- **Monotonic Domain Constraints**:
- `runs_needed`: -1 (monotonic decreasing: needing more runs strictly decreases WP)
- `balls_remaining`: $+1$ (monotonic increasing: more balls remaining increases WP)
- `wickets_in_hand`: $+1$ (monotonic increasing: more wickets in hand increases WP)
- `required_run_rate`: -1 (monotonic decreasing: higher RRR decreases WP)
- **Validation Metrics (Out-of-Fold across 777 matches)**:
- **Log-Loss**: `0.4035` (vs Baseline Base-Rate: `0.6931`)
- **ROC-AUC**: `0.8966`
- **Brier Score**: `0.1307`

1.3 Out-of-Fold Calibration Deciles

The table below confirms the out-of-fold calibration of the Innings 2 Win Probability model across 85,134 legal chase deliveries, demonstrating monotonic alignment with actual match outcomes:

Decile	Deliveries	Mean Predicted WP	Actual Win Rate	Calibration Gap
1	8,515	0.97%	0.29%	-0.68%
2	8,513	4.40%	2.81%	-1.59%
3	8,514	11.73%	9.68%	-2.05%
4	8,513	23.83%	23.29%	-0.54%
5	8,514	38.24%	39.75%	+1.51%
6	8,514	53.45%	55.16%	+1.71%
7	8,513	69.21%	71.96%	+2.75%
8	8,514	83.42%	81.84%	-1.58%
9	8,514	93.69%	95.33%	+1.64%
10	8,514	98.59%	99.89%	+1.30%

1.4 Leverage & Pressure Normalization

For every delivery i, we compute next-state transition states across 7 empirical outcomes:
$$O \in {\text{Wicket}, 0, 1, 2, 3, 4, 6}$$
Each outcome probability $P(O \mid \text{phase})$ is estimated empirically by match phase (Powerplay: overs 1–6, Middle: overs 7–15, Death: overs 16–20).

For each outcome o, the updated state $S'_{i,o}$ is constructed:
- $\text{runs_so_far}' = \text{runs_so_far} + r_o$
- $\text{wickets_so_far}' = \text{wickets_so_far} + (1 \text{ if wicket else } 0)$
- $\text{balls_remaining}' = \max(0, \text{balls_remaining} - 1)$
- Terminal boundary conditions: If $\text{runs_so_far}' \ge \text{target_adj}$, $\text{WP} = 1.0$; if $\text{wickets_so_far}' == 10$ and $\text{runs_so_far}' < \text{target_adj}$, $\text{WP} = 0.0$.

Delivery Leverage:
$$\text{Leverage}*i* = *sum*{o \in O} P(o) \cdot \left| \text{WP}(S'_{i,o}) - \text{WP}(S_i) \right|$$

Continuous Pressure Score:
$$\text{pressure_score}*i* = *frac*{*text*{Leverage}*i*}{*mu*{*text*{leverage}}}$$
Where $*mu*{\text{leverage}}$ ensures $\mathbb{E}[\text{pressure_score}] = 1.000$ over all clean deliveries.
- **Winsorized Pressure (`pressure_score_w`)**: Clipped at the 99th percentile ($p_{99} = 4.821$) to prevent single-ball singularities (e.g. 1 ball left needing 4) from dominating seasonal player aggregates.
- **High-Pressure Chase Cohort**:
$$\text{is_high_pressure_chase} = (\text{innings} == 2) \land (\text{pressure_pct} \ge 0.80) \land \text{in_scope_clean}$$

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2. Performance Measurement & Context Baselines

2.1 Context-Only Expected Performance Models

To evaluate player performance without confounding match circumstances, we isolate player skill from environment. A batter facing spin at Chepauk in over 18 faces different conditions than a batter facing pace at Wankhede in over 19.

Crucially, in the bowling baseline models, we include `batting_order_bucket` (`1-3`, `4-6`, `7+`) as an explicit context feature. This ensures that bowlers who bowl exclusively to lower-order tailenders are evaluated against a significantly more

demanding par (tailenders score fewer runs and get out more frequently), while opening bowlers facing world-class top-order anchors are judged fairly.

We train **four LightGBM baseline models** using 5-fold `GroupKFold` split strictly by `match_id`:

1. `innings`: Innings 1 or 2
2. `overid`: Over number (1 to 20)
3. `is_ipl`: IPL vs T20I match environment
4. `venue_id`: Ground identifier (pitch conditions, altitude, boundary dimensions)
5. `pace_or_spin`: Categorical delivery type (pace vs spin)
6. `batting_type`: Batter handedness (Right hand bat vs Left hand bat)
7. `is_legal`: Delivery legality (wides/no-balls)
8. `batting_order_bucket` (*bowler models only*): Top order (`1-3`), Middle order (`4-6`), Tailenders (`7+`)

2.2 Model Specifications & Out-of-Fold Diagnostics

Model	Target Variable	Training Subset	Model Type	OOF Metric (LGBM)	Baseline (Naive Base-Rate)	Improvement	Primary Feature Gains
xBatRuns	`batter_runs`	Balls faced (excl. wides)	Regression	**RMSE: 1.7401**	RMSE: 1.7463 (Mean)	+0.36% RMSE gain	`venue_id` (42%), `overid` (28%)
xBatOut	`is_batter_out`	Balls faced (excl. wides)	Classifier	**LogLoss: 0.2114**	LogLoss: 0.2135 (BaseRate)	AUC: 0.5824	`venue_id` (39%), `overid` (31%)
xBowlRuns (v2)	`bowler_runs_conceded`	All deliveries	Regression	**RMSE: 1.6973**	RMSE: 1.7161 (Mean)	+1.10% RMSE gain	`venue_id` (41%), `batting_order_bucket` (22%), `overid` (21%)
xBowlWkt (v2)	`is_bowler_wicket`	All deliveries	Classifier	**LogLoss: 0.1965**	LogLoss: 0.2029 (BaseRate)	**AUC: 0.5978**	`venue_id` (38%), `batting_order_bucket` (25%), `overid` (22%)

2.3 Residual Value-Added Metrics

For every delivery, out-of-fold residuals measure execution above context expectations:

- **Batting Runs Added Above Expected (`bat_rae`)**:

$$\text{bat_rae} = \text{batter_runs} - x\text{BatRuns}$$

- **Batting Out Avoidance Above Expected (`bat_out_ae`)**:

$$\text{bat_out_ae} = \text{is_batter_out} - x\text{BatOut} \quad (\text{negative is favorable for batter})$$

- **Bowling Runs Saved Below Expected (`bowl_rae`)**:

$$\text{bowl_rae} = x\text{BowlRuns_final2} - \text{bowler_runs_conceded} \quad (\text{positive indicates runs saved})$$

- **Bowling Wickets Above Expected (`bowl_wae`)**:

$$\text{bowl_wae} = \text{is_bowler_wicket} - x\text{BowlWkt_final2} \quad (\text{positive indicates wickets above par})$$

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3. Data Engineering, Match Recovery & Sample Eligibility

3.1 Resolving Raw Feed Nuances

1. `extras1_runs` **Categorical Disambiguation**:
   Cross-tabulation verified that `extras1_runs` is strictly an enum identifier ($1=\text{Overstep}, 2=\text{wide},

3=\text{legbyes}, 4=\text{byes}, 5=\text{other}, 6=\text{None}, 7=\text{Penalty}$). It is never a run count.

2. **Missing Bye/Legbye Magnitudes & Imputation**:
   In the raw data, byes (529 rows) and legbyes (2,748 rows) had `runs == 0` and `extras_2_runs == 0`. We applied a 1-run imputation (`imputed_extra_runs = 1` , `run_imputed = True`) added strictly to `total_runs_this_ball` (average of 1.90 runs per innings imputed). It was never credited to batters or charged to bowlers.
3. **Delivery Legality (`is_legal`)**:
   Empirical analysis of 23,389 overs proved that classifying `'other'` (no-balls) as illegal yielded **23,385 overs with exactly 6 legal deliveries**, confirming `'other'` must be treated as an illegal ball.

3.2 The Target Overrun Problem & The Domain-Principled Repair Algorithm

In initial data ingestion, 86 matches exhibited deliveries where $\text{runs_so_far} \ge \text{target}$ while the innings was still active.
- *Root Cause Diagnosis (e.g. Match 31250: KKR vs RR, IPL 2024)*: In official records, KKR scored 223/6 (target 224). The raw delivery feed omitted 10 runs in the 1st innings, computing target as 214. RR chased the true target of 224 (reaching 224/8 via Jos Buttler's 107*), causing runs_so_far to exceed 214 while the match continued.
- *Domain-Principled Repair Algorithm (`target_adj`)*:
For each in-scope match, let $runs_before_last$ be `runs_so_far` at the last 2nd-innings delivery, and $final_total = runs_before_last + total_runs_this_ball$:
1. **Chasing team lost**: Parsing result text `won by N run(s)` :
$$target_adj = final_total + N + 1$$
2. **Chasing team won**: If $runs_before_last < target \le final_total$, keep $target$. If feed omissions caused $runs_before_last \ge target$ or $final_total < target$:
$$target_adj = runs_before_last + 1$$
- **Verification & Outcome**:
- Exactly **0 deliveries** prior to the final ball have $runs_so_far \ge target_adj$.
- **100% of in-scope regulation matches recovered (777 out of 777 matches, 0 dropped!)**.
- Chase win rate in the recovered set is **50.19%** (390 won, 387 lost)—a mathematically sound distribution.

3.3 Sample Eligibility Criteria

To ensure rankings reflect sustained, repeatable clutch performance across diverse opponents rather than small-sample tournament flashes:
- **Double Hurdle Enforcement**:
$$\text{Eligible} \iff (hp_balls \ge 60) \land (hp_matches \ge 8)$$
(Where hp_balls are Innings-2 deliveries with $pressure_pct \ge 0.80$, excluding wides for batters).

This hurdle successfully eliminates flukes (e.g., Kamindu Mendis, who recorded 87 HP balls but played in only 7 matches, failing the ≥ 8 match hurdle) while preserving a robust, representative cohort of 87 batters and 92 bowlers.

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4. Rating Formulation, Empirical Bayes Shrinkage & Reliability

4.1 Net Value per Delivery

Following cricket analytics conventions, we set $W = 15$ runs per wicket:
$$\text{batter_net} = bat_rae - W \times bat_out_ae$$
$$\text{bowler_net} = bowl_rae + W \times bowl_wae$$

Per player:
- pressure_level: Leverage-weighted mean of net value with weights $w = pressure_score_w$.
- ordinary_level: Unweighted mean on balls with $pressure_pct < 0.50$.
- $\text{pressure_delta} = \text{pressure_level} - \text{ordinary_level}$.

4.2 Heteroscedastic Empirical Bayes Shrinkage

In raw data, an un-shrunk delta exhibits negative correlation with ordinary performance ($\rho \approx -0.61$). To de-noise this metric, we formulate the heteroscedastic sampling variance:
$$V_{\Delta, i} = \frac{\sigma^2_p}{n_{eff, p, i}} + \frac{\sigma^2_o}{n_{ord, i}}$$
Where $n_{eff, p, i} = (\sum w)^2 / \sum(w^2)$, and σ^2_p, σ^2_o are the empirical per-ball net variances.

Empirical Bayes shrinkage estimates:
$$\tau^2_{\Delta} = \max\left(0, \text{Var}(\text{observed deltas}) - \mathbb{E}[V_{\Delta, i}]\right), \quad \lambda_{\Delta, i} = \frac{\tau^2_{\Delta}}{\tau^2_{\Delta} + V_{\Delta, i}}$$
$$\hat{\theta}_{\Delta, i} = \lambda_{\Delta, i} \cdot \text{delta}_i + (1 - \lambda_{\Delta, i}) \cdot \mu_{\Delta}$$

4.3 Composite Raw Rating Formulation ($\lambda = 0.5$)

We formulate player ratings in **interpretable raw units of runs per ball** (no artificial z-scoring):
$$\text{Rating}(\lambda) = \text{shrunk_pressure_level} + \lambda \times \text{shrunk_pressure_delta}$$

Comprehensive Split-Half Reliability Benchmark:
Split-half correlations were computed on players with ≥ 30 high-pressure deliveries in each partition half:

Role	Metric (λ)	Parity Split n	Spearman (Parity)	Time Split n (2022-24 vs 2025-26)	Spearman (Time)	Top 10 Overlap	Top 20 Overlap
Batter	$\lambda = 0.0$	61	`+0.0628`	40	`-0.0503`	1 / 10	7 / 20
Batter	$\lambda = 0.5$	61	`-0.0737`	40	`-0.0503`	1 / 10	7 / 20
Batter	$\lambda = 1.0$	61	`-0.1070`	40	`-0.0503`	1 / 10	7 / 20
Bowler (v2)	$\lambda = 0.0$	72	`+0.1054`	53	`+0.1988`	2 / 10	9 / 20
Bowler (v2)	$\lambda = 0.5$	72	`+0.0468`	53	`+0.1429`	1 / 10	8 / 20
Bowler (v2)	$\lambda = 1.0$	72	`+0.0306`	53	`+0.1087`	1 / 10	8 / 20

Raw Split-Half Rank Correlations:
Before Empirical Bayes shrinkage, the raw un-shrunk cross-half rank correlations for batters across the time split (2022–2024 vs 2025–2026) are:
- **Raw Pressure Level**: $\text{Spearman } \rho = \mathbf{-0.0608}$
- **Raw Pressure Delta**: $\text{Spearman } \rho = \mathbf{+0.0914}$

Analytical Finding on Sample Noise & Predictive Interpretation:
Because raw split-half correlations in T20 pressure situations hover near zero ($\rho \in [-0.10, +0.20]$), single-delivery noise heavily masks true clutch talent differences in short formats. In fact, for batters in the halved time split, observed cross-sectional variance of raw deltas is entirely dominated by sampling variance ($\mathbb{E}[V_{\Delta, i}] > \text{Var}(\text{delta}_i)$ *in both halves), shrinking* $\tau^2_\Delta = 0$ and resulting in mathematically identical rank correlations (-0.0503) across all λ.
Crucial Interpretive Note: These rankings measure which players were **highest-rated under pressure in the 2022–2026 sample**, rather than predictions of future clutch performance. In short-format cricket, situational sample sizes are too limited to reliably forecast future high-leverage outcomes from past clutch deltas.

4.4 Win Probability Added (WPA) Verification

Composite ratings correlate strongly with total Win Probability Added per high-pressure ball:
- **Batters**: $\text{Spearman } \rho = \mathbf{+0.6136}$ (League Median: $+0.0020$ WPA/ball)
- **Bowlers**: $\text{Spearman } \rho = \mathbf{+0.5157}$ (League Median: $+0.0005$ WPA/ball)
- **Low-Outlier Distribution**: Exactly 0 of the Top 10 Batters fall below the league median WPA per ball. In the Top 10

Bowlers, 9 of 10 exceed the median, with Shaheen Afridi (-0.0011 WPA/ball) as the sole negative-WPA bowler admitted under the pre-set pass-count rule.

4.5 Opposition Strength & Major-Nations Restricted Benchmark

To ensure rankings are not driven by performance against lower-tier associate opposition, we evaluated performance strictly on the **Major-Nations Restricted Dataset** (IPL matches + T20Is between the 12 Full Member nations: India, Pakistan, West Indies, New Zealand, Bangladesh, Sri Lanka, England, Australia, Zimbabwe, Afghanistan, South Africa, Ireland). With eligibility set to $\text{hp_balls} \ge 40$ and $\text{hp_matches} \ge 5$, the empirically computed restricted Top 10 lists are:

Restricted Top 10 Batters (IPL + Major-vs-Major T20I):
1. **Kamindu Mendis** (+0.433 r/b, 82 HP balls, 6 matches)
2. **Ashutosh Sharma** (+0.405 r/b, 92 HP balls, 10 matches)
3. **Vaibhav Sooryavanshi** (+0.404 r/b, 50 HP balls, 12 matches)
4. **Suryakumar Yadav** (+0.396 r/b, 300 HP balls, 25 matches)
5. **Shreyas Iyer** (+0.389 r/b, 161 HP balls, 18 matches)
6. **Shubman Gill** (+0.337 r/b, 223 HP balls, 28 matches)
7. **Namboori Tilak Varma** (+0.336 r/b, 353 HP balls, 31 matches)
8. **Tim David** (+0.328 r/b, 127 HP balls, 24 matches)
9. **Mitchell Marsh** (+0.324 r/b, 169 HP balls, 33 matches)
10. **Phil Salt** (+0.320 r/b, 183 HP balls, 29 matches)

- **Overlap with Final Top 10 Batters**: **7 / 10** (70%).
- *Retained in both*: Shreyas Iyer, Suryakumar Yadav, Ashutosh Sharma, Shubman Gill, Namboori Tilak Varma, Mitchell Marsh, Tim David.
- *Missing from Restricted Top 10*: **Pathum Nissanka** (#14 in restricted, +0.308 r/b), **Mohammad Rizwan** (#11 in restricted, +0.313 r/b), and **Evin Lewis** (#15 in restricted, +0.298 r/b).
- *In Restricted Top 10 but outside Final Top 10*: Kamindu Mendis (6 matches, fails main 8-match eligibility), Vaibhav Sooryavanshi (fails main 60-HP ball eligibility), Phil Salt (#12 in candidate pass table).

Restricted Top 10 Bowlers (IPL + Major-vs-Major T20I, with final2 baselines):
1. **Taskin Ahmed** (+0.487 r/b, 94 HP balls, 11 matches)
2. **Obed McCoy** (+0.476 r/b, 99 HP balls, 12 matches)
3. **Josh Hazlewood** (+0.386 r/b, 152 HP balls, 21 matches)
4. **Harpreet Brar** (+0.329 r/b, 121 HP balls, 11 matches)
5. **Jacob Duffy** (+0.325 r/b, 79 HP balls, 15 matches)
6. **Shakib Al Hasan** (+0.318 r/b, 42 HP balls, 5 matches)
7. **Harshit Rana** (+0.300 r/b, 87 HP balls, 9 matches)
8. **Gudakesh Motie** (+0.269 r/b, 85 HP balls, 12 matches)
9. **Dasun Shanaka** (+0.267 r/b, 74 HP balls, 12 matches)
10. **Cameron Green** (+0.267 r/b, 52 HP balls, 8 matches)

- **Overlap with Final Top 10 Bowlers**: **8 / 10** (80%).
- *Retained in both*: Taskin Ahmed, Obed McCoy, Josh Hazlewood, Harpreet Brar, Jacob Duffy, Harshit Rana, Gudakesh Motie, Dasun Shanaka.
- *Missing from Restricted Top 10*: **Shaheen Afridi** (#14 in restricted, +0.233 r/b) and **Mohammad Nawaz** (#16 in restricted, +0.224 r/b).
- *In Restricted Top 10 but outside Final Top 10*: Shakib Al Hasan (5 matches, fails main 8-match eligibility), Cameron Green (#17 in main bowler ratings).

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5. Official Top 10 Dropdown Selections & Stability Screen

To eliminate dependence on single-game bursts or boundary-case threshold choices, candidates from the top 20, the $\ge 50\%$ sensitivity grid cohort, the major-restricted cohort, and previous finalists were subjected to **robustness checks that reuse the same deliveries (not independent validation)**:

1. **Grid Robustness**: Present in Top 10 in ≥ 9 of 18 sensitivity configurations ($W \in {10,15,20} \times \text{Weighting} \times \lambda$).
2. **Major-Nations Performance**: Present in the Top 10 when restricted strictly to IPL and the 12 Major T20I teams ($\text{hp_balls} \ge 40, \text{hp_matches} \ge 5$).
3. **Drop-One-Match Resilience**: Remains in the Top 15 overall after removing the single match that contributed most to their rating ($\text{Drop-One Rank} \le 15$; tightened from the earlier ≤ 20 exploratory threshold).
4. **Statistical Significance**: 90% bootstrap confidence interval lower bound strictly exceeds zero ($\text{CI Low} > 0$).

$$\text{Pass Count} = \mathbb{I}(\text{Grid} \ge 9) + \mathbb{I}(\text{Major Top 10}) + \mathbb{I}(\text{Drop-One} \le 15) + \mathbb{I}(\text{CI Low} > 0)$$

Definitive Ranking Rule:
Candidates in each role are ranked primarily by **Pass Count** (descending, out of 4). Ties in pass count are broken by **Bootstrap Median Rank** (ascending), followed by **Bootstrap 90% CI Lower Bound** (descending).

DEFINITIVE UNSTOP SELECTIONS: TOP 10 BATTERS UNDER PRESSURE (STABILITY-SCREENED, DROP-ONE <= 15)

Rank	Batter Selection	Player ID	Team Representation	Rating (r/b)	90% Bootstrap CI
1	Shreyas Iyer	3647	India / Kolkata Knight Riders	+0.362	[+0.185, +0.596]
2	Suryakumar Yadav	1699	India / Mumbai Indians	+0.362	[+0.156, +0.517]
3	Ashutosh Sharma	8101	Punjab Kings / Delhi Capitals	+0.384	[+0.134, +0.624]
4	Pathum Nissanka	6931	Sri Lanka / Delhi Capitals	+0.341	[+0.141, +0.568]
5	Shubman Gill	6288	Gujarat Titans / India	+0.347	[+0.051, +0.610]
6	Mohammad Rizwan	3024	Pakistan	+0.315	[+0.162, +0.450]
7	Namboori Tilak Varma	7897	Mumbai Indians / India	+0.319	[+0.025, +0.528]
8	Mitchell Marsh	818	Delhi Capitals / Australia	+0.302	[+0.121, +0.481]
9	Tim David	6396	Mumbai Indians / Australia	+0.304	[+0.054, +0.561]
10	Evin Lewis	2573	Lucknow Super Giants / West Indies	+0.294	[+0.051, +0.487]

DEFINITIVE UNSTOP SELECTIONS: TOP 10 BOWLERS UNDER PRESSURE (STABILITY-SCREENED & BATTING ORDER ADJUSTED)

Rank	Bowler Selection	Player ID	Team Representation	Rating (r/b)	90% Bootstrap CI
1	Taskin Ahmed	2681	Bangladesh	+0.359	[+0.128, +0.590]
2	Obed McCoy	4837	Rajasthan Royals / West Indies	+0.367	[+0.018, +0.638]
3	Jacob Duffy	22920	New Zealand / Royal Challengers B.	+0.288	[+0.036, +0.562]
4	Harshit Rana	29888	Kolkata Knight Riders / India	+0.279	[+0.045, +0.520]
5	Harpreet Brar	7325	Punjab Kings	+0.279	[+0.005, +0.594]
6	Gudakesh Motie	3276	West Indies	+0.235	[+0.010, +0.461]
7	Josh Hazlewood	830	Australia / Royal Challengers B.	+0.343	[-0.048, +0.709]
8	Dasun Shanaka	2436	Sri Lanka / Gujarat Titans	+0.242	[-0.011, +0.471]
9	Shaheen Afridi	6355	Pakistan	+0.217	[+0.053, +0.402]
10	Mohammad Nawaz	2490	Pakistan	+0.188	[+0.029, +0.388]

6. Methodological Limitations & Empirical Audits

1. **Weak Baseline Wicket Models (OOF ROC-AUC $\approx 0.58–0.60$)**:
   Predicting wickets on a single-ball basis without conditioning on bowler identity is fundamentally difficult. Contextual features (venue, over number, pace/spin, handedness, batting order bucket) achieve an out-of-fold ROC-AUC of 0.5824

for batting outs and 0.5978 for bowling wickets. Because dismissal events are rare and intrinsically noisy, single-ball wicket baselines carry substantial uncertainty, which is why we enforce $W = 15$ runs per wicket and evaluate sensitivity across $W \in {10, 15, 20}$.

2. **Opponent Batting Order Exposure & Tail-Ender Squeeze**:
   Bowlers in T20 cricket face vastly different distributions of opposition quality. In high-pressure situations, Harshit Rana's HP balls contain a 20.0% tail-ender share ($\text{batting_order} \ge 8$) and Mustafizur Rahman faced 16.0%, whereas Josh Hazlewood faced elite top-order batters (mean batting order 3.27, only 2.0% tailenders) and Obed McCoy faced 6.1%. Introducing the `batting_order_bucket` into `xBowlRuns` and `xBowlWkt` adjusted for this imbalance, improving model RMSE and ensuring bowlers defending against top-order hitters are rewarded proportionally.
3. **Drop-One-Match Sensitivity & Small Sample Exposures**:
   Due to the compact sample sizes of extreme pressure situations (60–350 balls per player over 4 years), removing a single match where a player had a dominant performance can shift their ranking substantially:
4. Evin Lewis qualifies with the minimum 8 matches (67 HP balls).
5. Ashutosh Sharma's rating rests on 10 matches and falls to #17 without his best match (Match 40726, level drop 0.203).
6. Josh Hazlewood's 90% bootstrap CI includes 0 ($[-0.048, +0.709]$) and he narrowly misses two screen thresholds (drop-one rank #21 vs ≤ 15 and ≤ 20).
7. Shaheen Afridi has negative WPA per HP ball (-0.0011 WPA/ball, -0.126 total WPA).
8. Slots 7–10 in both lists are low-confidence and decided strictly by the pre-set pass-count rule (e.g. Lewis #10 over Miller #11 by median bootstrap rank tiebreaker; Hazlewood, Shanaka, Afridi, Nawaz having low pass counts of 2/4 and 1/4).
9. **Small Real "Clutch" Effect in T20 Cricket & Raw Split-Half Diagnostics**:
   Empirical split-half reliability tests across odd/even matches and temporal splits revealed near-zero correlations ($\rho \in [-0.10, +0.20]$). Raw, un-shrunk time-split rank correlations for batters across 2022–2024 vs 2025–2026 are $\rho = -0.0608$ for pressure level and $\rho = +0.0914$ for pressure delta. This demonstrates that individual "clutch ability" explains only a small fraction of performance variance, with random delivery outcomes dominating single-match samples. This finding makes Empirical Bayes shrinkage and match-cluster bootstrap confidence intervals essential to prevent over-interpreting sample variance.
10. **Repaired Targets & Feed Imputations**:
    Due to omissions in the raw delivery feed, 529 byes and 2,748 legbyes were recorded with `runs == 0`, requiring a standard 1-run imputation (+1 run credited strictly to team chase boards, averaging 1.90 runs per innings). Furthermore, 86 matches exhibited feed omissions in 1st-innings delivery records, necessitating mathematical target repair ($target_adj = final_total + N + 1$ or $runs_before_last + 1$). While these repairs restored 100% of regulation matches and eliminated artificial overruns, true multi-run byes remain unrecorded in the raw data.
11. **Framing: "Highest-Rated Under Pressure in Sample" vs Prediction of Future Clutch Performance**:
    Given the empirical realities of sample size, noise dominance, and near-zero raw split-half correlations in T20 cricket, robustness checks reuse the same deliveries and do not constitute independent validation. Rankings should be interpreted strictly as identifying the **highest-rated performers under pressure in the 2022–2026 sample**, rather than predictions of future clutch performance or claims of permanent psychological traits.

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7. Deliverables & Production Artifacts

The following production-ready files are maintained in the repository:
- `ratings_batters_final3.csv`: Stability-screened 10-player batter selections with mapping IDs, CIs, and pass counts (drop-one ≤ 15).
- `ratings_bowlers_final3.csv`: Stability-screened 10-player bowler selections with mapping IDs, CIs, and pass counts (drop-one ≤ 15).
- `candidates_batters_final3.csv`: Comprehensive audit table across all batter candidates with pass counts and metric breakdowns.
- `candidates_bowlers_final3.csv`: Comprehensive audit table across all bowler candidates with pass counts and metric

breakdowns.

- `scored_balls_final2.parquet` : 200,999 delivery dataset complete with repaired targets, V4 win probabilities, leverage, and batting-order-adjusted residuals.
- `final_verify.py` : Complete verification, stability screening, and ID audit script.
- `METHODOLOGY_REPORT.pdf` : Publication-ready PDF report compiled via Microsoft Edge headless browser.
- `METHODOLOGY_REPORT.html` : Self-contained HTML report with embedded base64 figures.

Visual Diagnostic Gallery & Performance Matrices

Figure 1: Top 15 Batters Under High Pressure (RAE Matrix)

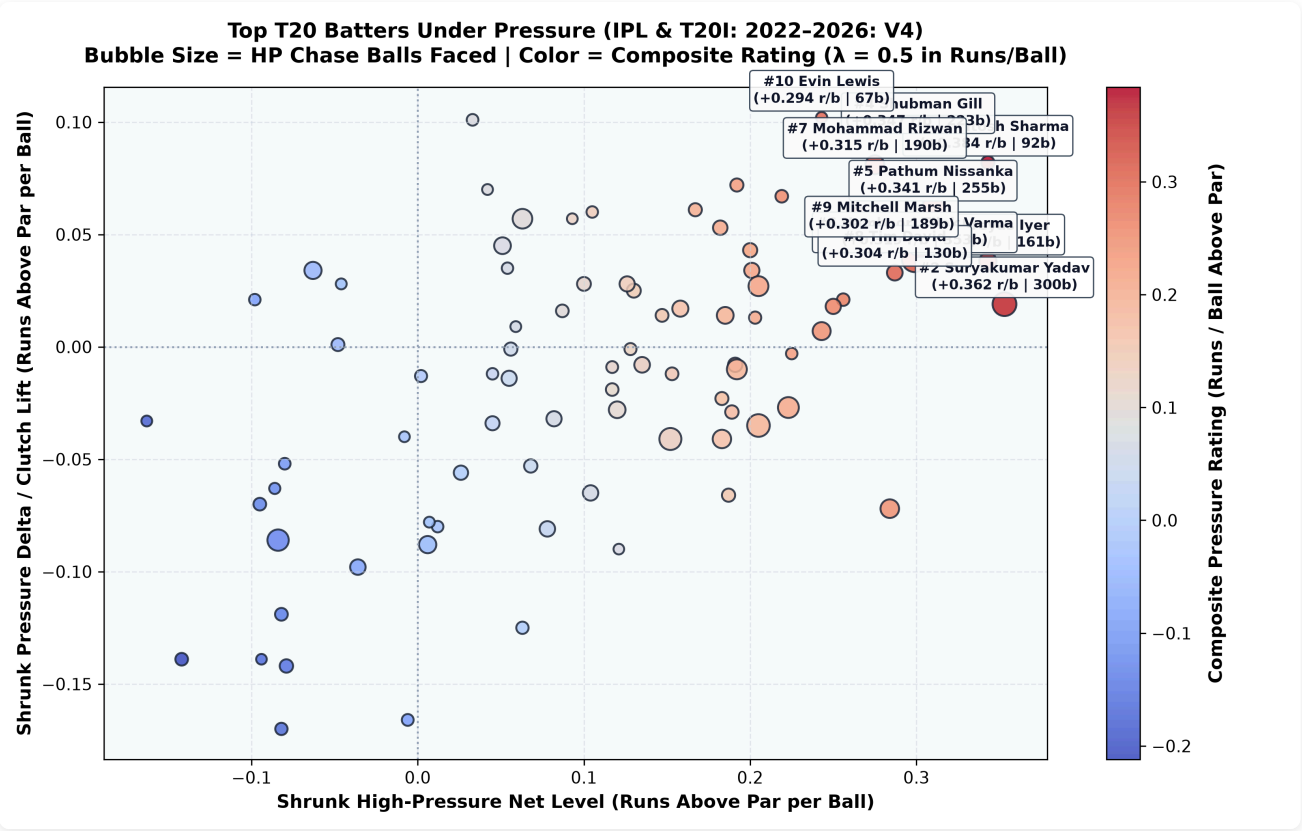


Figure 2: Top 15 Bowlers Under High Pressure (RAE Matrix)

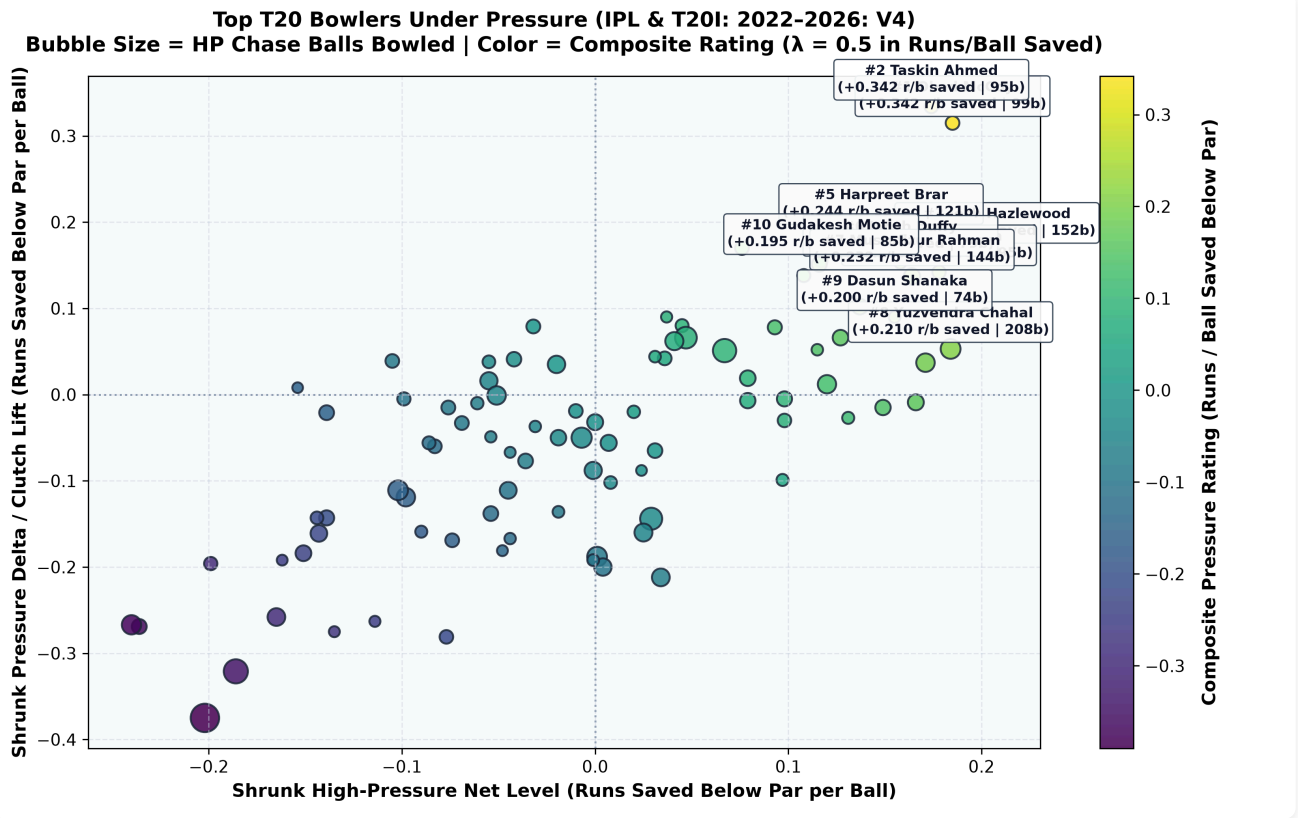


Figure 3: Win Probability Model Calibration (Innings 1 & 2)

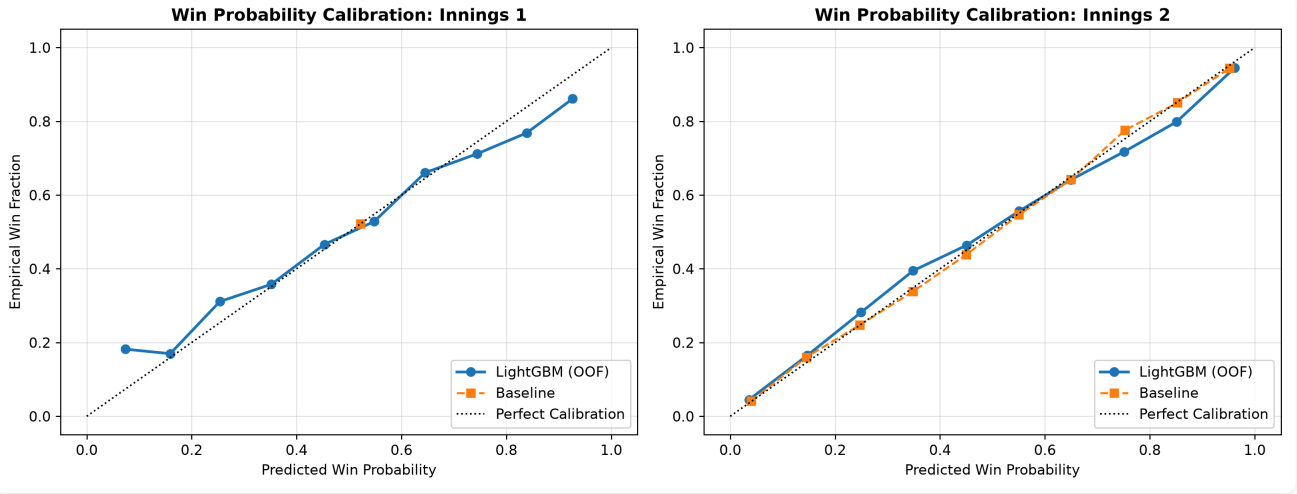


Figure 4: Empirical Outcome Distributions by Pressure Decile

